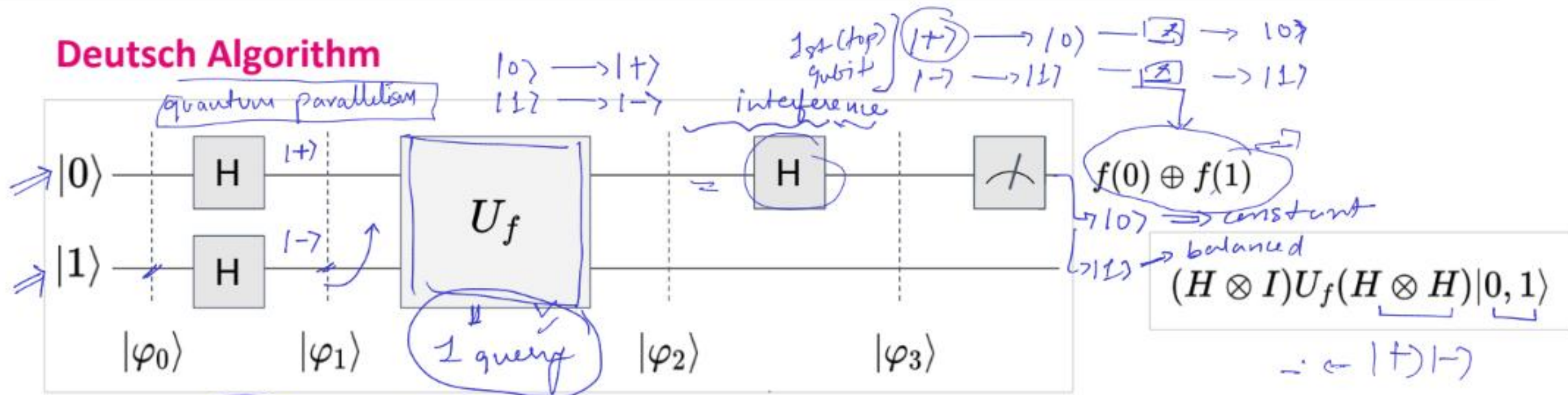


Deutsch Algorithm



$$|\varphi_0\rangle = |0,1\rangle$$

$$|\varphi_1\rangle = \left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right] \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right] = \frac{+|0,0\rangle - |0,1\rangle + |1,0\rangle - |1,1\rangle}{2}$$

$$|\varphi_2\rangle = \left[\frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}} \right] \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]$$

$$|\varphi_2\rangle = \begin{cases} (\pm 1) \left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right] \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right], & \text{if } f \text{ is constant} \\ (\pm 1) \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right] \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right], & \text{if } f \text{ is balanced} \end{cases}$$

$$|\varphi_3\rangle = \begin{cases} (\pm 1)|0\rangle \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right], & \text{if } f \text{ is constant} \\ (\pm 1)|1\rangle \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right], & \text{if } f \text{ is balanced} \end{cases}$$

quantum parallelism

phase effect

interference

$$|+\rangle|-\rangle \xrightarrow{U_f} (-1)^{f(x)}|x\rangle|-\rangle$$

$$f(0) = f(1) \rightarrow f(0) = f(1) = 0 \rightarrow +(|0\rangle + |1\rangle) \rightarrow \pm 1(|0\rangle + |1\rangle)$$

$$f(0) \neq f(1) \rightarrow f(0) = 0, f(1) = 1 \rightarrow -|0\rangle - |1\rangle \rightarrow -(|0\rangle + |1\rangle)$$

constructive interference

destructive interference

$$\left. \begin{matrix} f(0)=0 \\ f(1)=1 \end{matrix} \right\} -$$

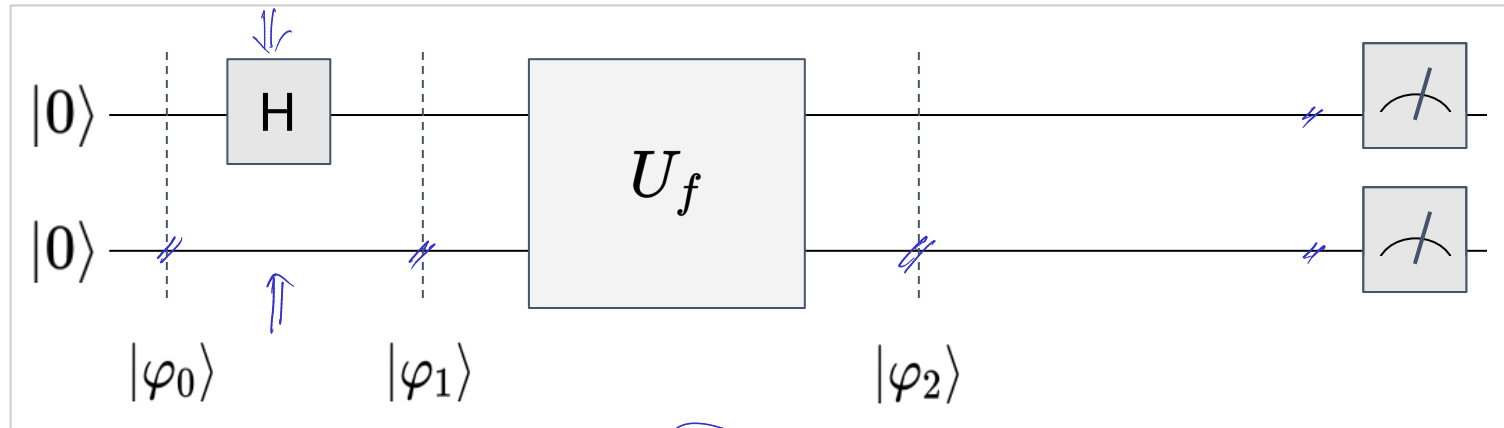
$$\left. \begin{matrix} f(0)=1 \\ f(1)=0 \end{matrix} \right\} -$$

$$|\varphi_2\rangle = \begin{cases} \pm |+\rangle|-\rangle \Rightarrow \text{const} \\ \pm |-\rangle|-\rangle \Rightarrow \text{bal} \end{cases}$$

$$|+\rangle \Rightarrow |0\rangle \text{ or } |1\rangle$$

$$|-\rangle \Rightarrow |0\rangle \text{ or } |1\rangle$$

Building a quantum circuit to solve Deutsch Problem (1)



$$|\varphi_0\rangle = |0\rangle \otimes |0\rangle = |0,0\rangle \quad |x\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

$$|\varphi_1\rangle = \left[\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right] \otimes |0\rangle = \frac{|0,0\rangle + |1,0\rangle}{\sqrt{2}} \xrightarrow{U_f}$$

$$|\varphi_2\rangle = \frac{|0, f(0)\rangle + |1, f(1)\rangle}{\sqrt{2}}$$

$$[f(0), f(1)]$$

$$\frac{f(0) \oplus f(1)}{1 \oplus 0} \quad ??$$

$$f: \{0,1\} \rightarrow \{0,1\}$$

$$[f(0) \oplus f(1)] \text{ classical}$$

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0, f(0)\rangle + |1, f(1)\rangle) \xrightarrow{1 \oplus 1} |x\rangle = |+\rangle$$

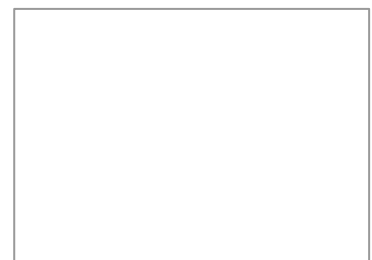
$$U_f(H \otimes I)(|0\rangle \otimes |0\rangle)$$

$|\varphi_2\rangle \quad |\varphi_1\rangle \quad |\varphi_0\rangle$

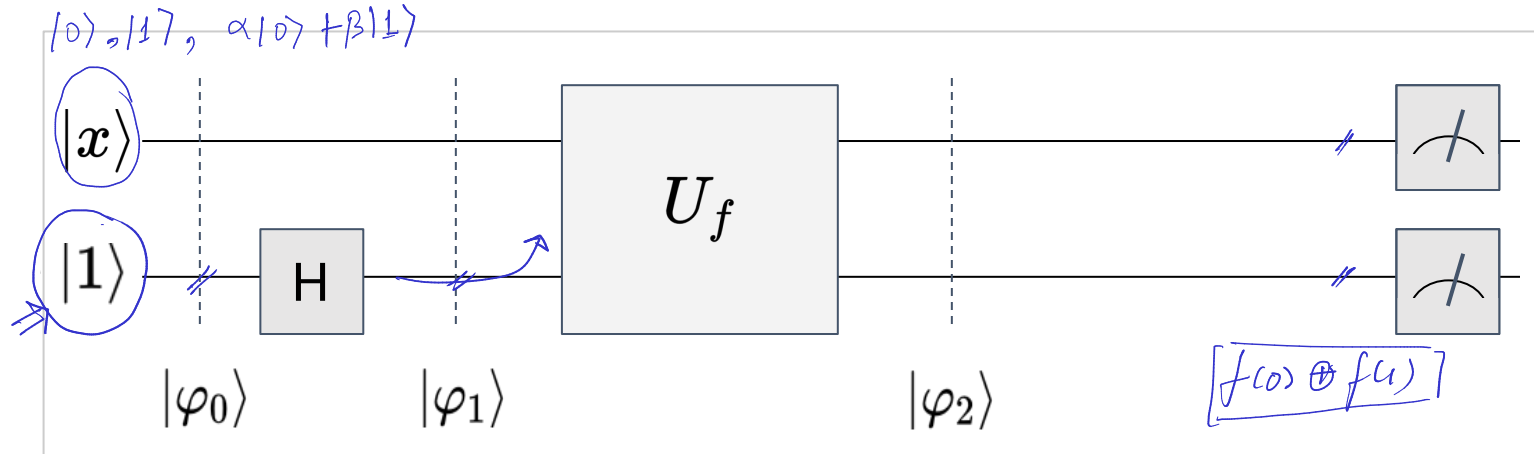
$$|x, 0\rangle \rightarrow |x, f(x)\rangle$$

$$\left| \frac{x_1 + x_2}{\sqrt{2}}, 0 \right\rangle \rightarrow$$

quantum parallelism



Building a quantum circuit to solve Deutsch Problem (2)



$$1/2 \rightarrow 1-7$$

$$\frac{1/0 - 1/1}{\sqrt{2}}$$

$$U_f(I \otimes H)|x, 1\rangle$$

$$|\varphi_0\rangle = |x, 1\rangle$$

$$|\varphi_1\rangle = |x\rangle \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right] = \left(\frac{|x, 0\rangle - |x, 1\rangle}{\sqrt{2}} \right)$$

$$|x, y\rangle \rightarrow |x, y \oplus f(x)\rangle$$

$$|x, -\rangle \rightarrow |x, - \oplus f(x)\rangle$$

$$|\varphi_2\rangle = |x\rangle \otimes \left[\frac{|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}} \right] = |x\rangle \otimes \left[\frac{|f(x)\rangle - |\overline{f(x)}\rangle}{\sqrt{2}} \right]$$

$$f: \{0, 1\} \rightarrow \{0, 1\}$$

phase effect

$$|\varphi_2\rangle = \begin{cases} |x\rangle \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right], & \text{if } f(x) = 0 \\ |x\rangle \otimes \left[\frac{|1\rangle - |0\rangle}{\sqrt{2}} \right], & \text{if } f(x) = 1 \end{cases}$$

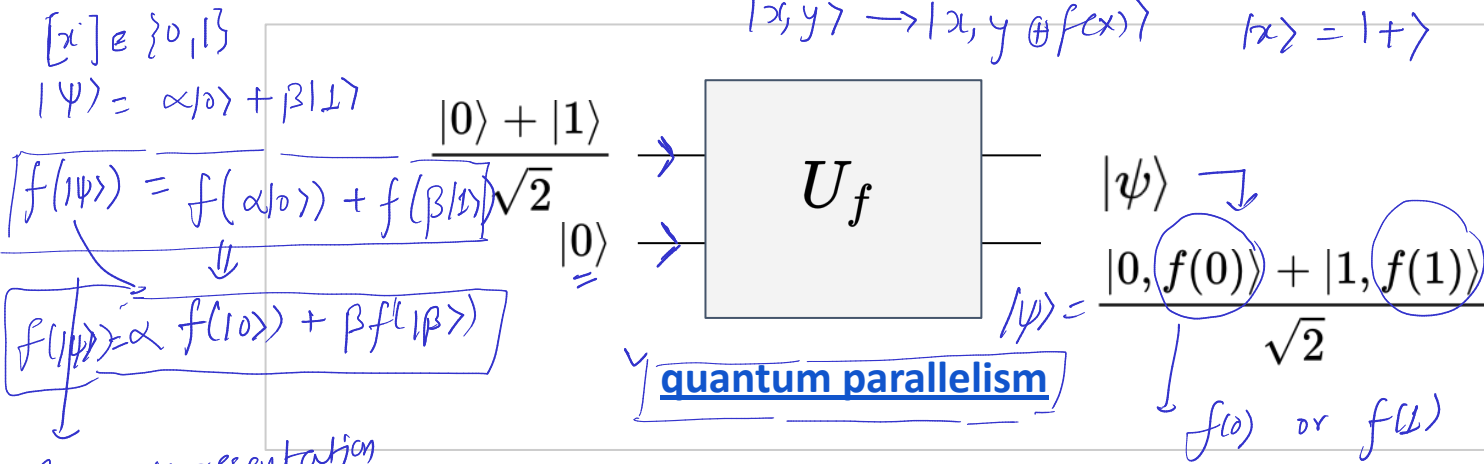
$$f(x) = 0 \rightarrow |0\rangle - |1\rangle$$

$$f(x) = 1 \rightarrow |1\rangle - |0\rangle$$

$$|\varphi_2\rangle = (-1)^{f(x)} |x\rangle \otimes \left[\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right]$$

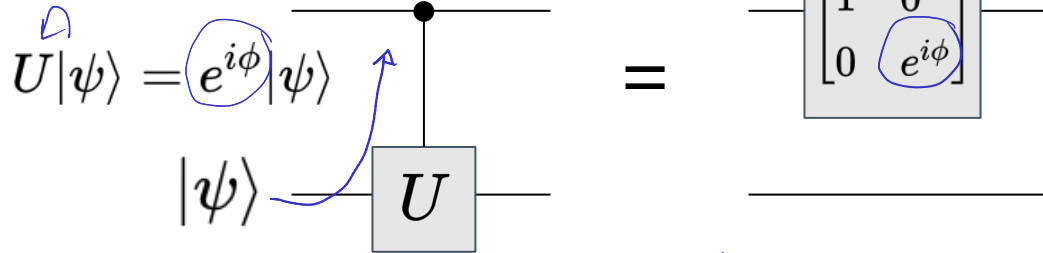
$$|\varphi_2\rangle = (-1)^{f(x)} |x\rangle |-\rangle$$

Quantum phenomenon as a resource in algorithm design



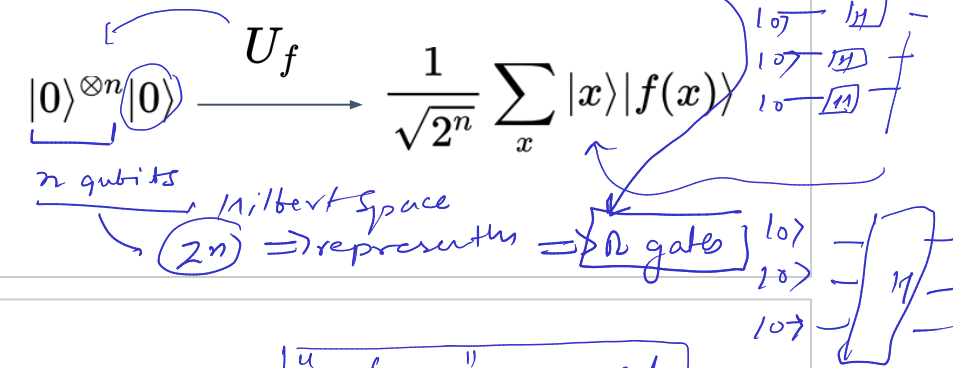
State representation
(linear superposition)

Phase Kick-back (trick)



QFT/shor's ...

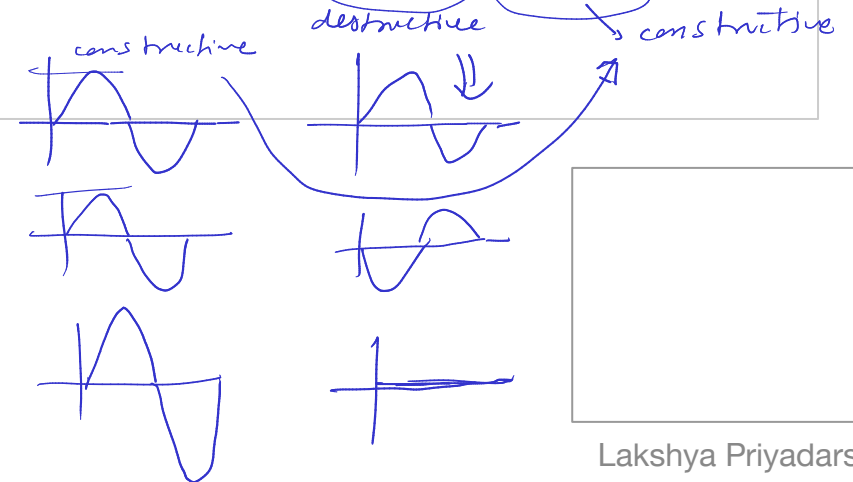
Walsh-Hadamard Transform



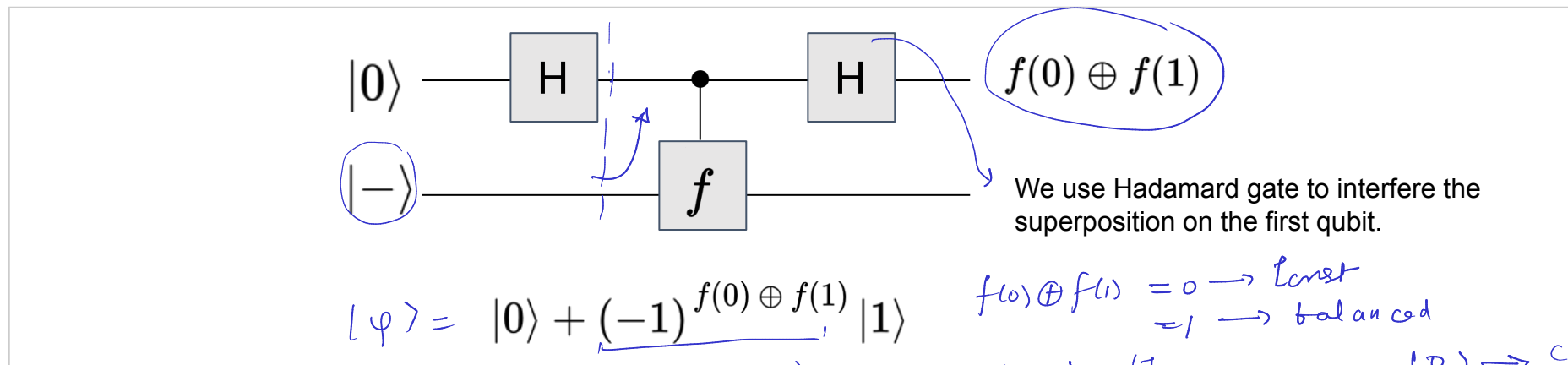
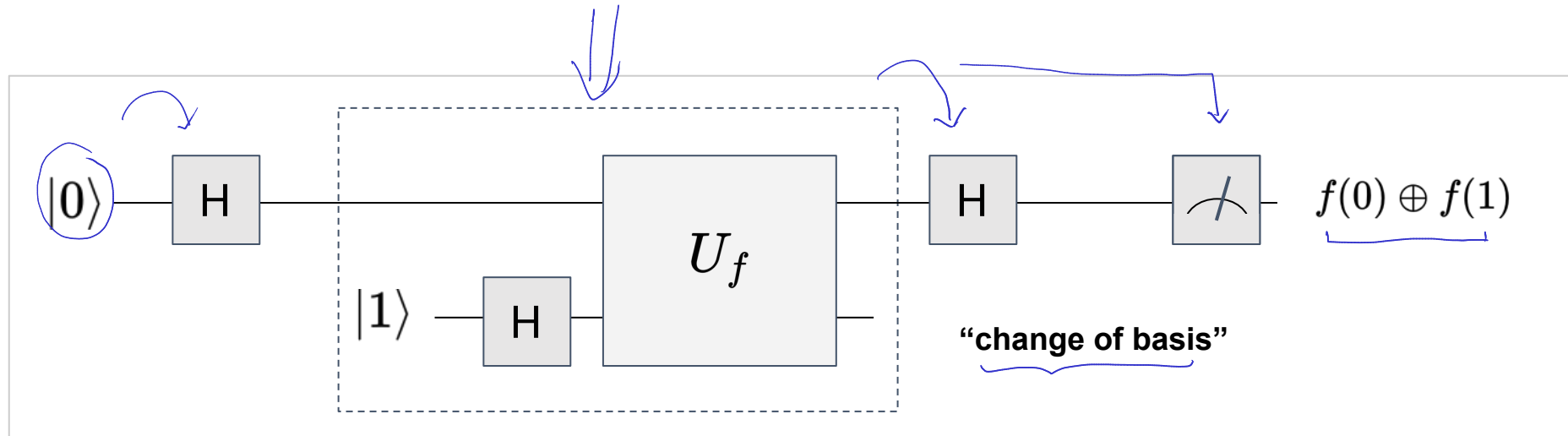
Interference

"enforce" $\rightarrow$ correct
 "discard" $\rightarrow$ don't require

Deutsch algorithm uses quantum interference to determine the property of the function being balanced or constant.



Deutsch Algorithm



$$|\varphi\rangle = |0\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle$$

$f(0) \oplus f(1) = 0 \rightarrow \text{const}$
 $= 1 \rightarrow \text{balanced}$

$|0\rangle + |1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \rightarrow \text{const}$
 $|0\rangle - |1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \rightarrow \text{bal}$

We prepare a state in superposition, perform some operations, and then use interference to discern some global property of the state.

2 queries $\rightarrow$ 1 query
(classical) (quantum)

$f: \{0,1\} \rightarrow \{0,1\}$

$$f(0) \oplus f(1)$$

We did not create any new information, we have only rearranged the information.

classical
1 query

PROMISE PROBLEM

The quantum algorithm exploited the structure of the problem to solve it faster than the classical algorithm.

superposition states
& using interference

